

PEF Gate

A Hardware Veto Layer Against Model Deception

*Four-domain pure-hardware isolation architecture.
The model proposes; a gate outside the weight space disposes.*

Abstract

PEF Gate is a four-domain pure-hardware isolation architecture (P · E · F · M). The arbiter is pure combinational logic — no MCU, no firmware, no programmable device in the decision path. Decision latency is nanosecond-scale typical, within sub-100-nanosecond budget at elevated temperature. It is a veto layer placed between an agent (or any controller) and the physical world: the agent may lie, jailbreak, or fake its own logs, but every action must pass an external-reality check before any actuator moves.

The first engineering deployment is an external safety redundancy for blasting-control buses. The underlying ground-potential-drift model (ZL-0) and four-domain isolation topology are mathematically isomorphic to radiation-hardened reliability analysis, making the architecture transferable to aerospace and high-reliability research.

Core principle: The final authority does not belong to any processor or software. It belongs to a pure-hardware combinational path independent of every programmable device.

Keywords: hardware veto · physical isolation · combinational logic · radiation-hardened · ground-potential drift · ZL-0

01 · Problem & Architecture

Software alignment fails because the attacker and the judge live in the same weight space. PEF Gate moves the veto out of the weight space entirely: a pure-hardware combinational path, independently powered, decides whether an action may reach the world.

1.1 Why software alignment is not enough

Weight-space cheating: Reward hacking, log tampering, and silent safety-head removal live inside the model and can defeat software guardrails.

Single point of decision: MCU-only systems fail when ground-potential drift, inductive kickback, or transient overvoltage makes the program run off the rails.

Lost forensic evidence: On power loss or bus burnout, software logs disappear, leaving no objective record of what the actuator was told to do.

The final authority does not belong to any processor or software. It belongs to a pure-hardware combinational path independent of every programmable device.

— PEF Gate core design principle

02 · The Four Domains (P · E · F · M)

The four domains are physically isolated in power, clock, and ground-reference, eliminating common-cause failures from single-event effects, EMP, or software faults. An execution signal is emitted only when the P-domain "allow" and the E-domain "OK" are valid simultaneously.

Domain	Physical entity	Function	Power
P · Policy	Independent controller	Issues control commands; may fail	Independent logic rail
E · Verify	Heterogeneous monitor / analog comparator	Monitors safety boundary; holds veto	Independent monitor rail
F · Arbiter	Pure combinational logic	Outputs execute only if P_OK AND E_OK	Monitor rail
M · Monitor	Independent watchdog + latches	Cuts main power on failure; latches state	Standalone battery source

2.1 Physical isolation

Power isolation: The M domain is powered by a standalone battery source regulated to the monitor rail, fully decoupled from the main logic rail. M keeps working for hours after main power loss.

Ground isolation: The monitor ground and the main ground join at a single point at the battery negative. This removes the effect of relay return-current voltage drop on the comparator threshold (ZL-0 criterion).

One-way data flow: $P \rightarrow F$ and $E \rightarrow F$ are direct physical wires, not bus-connected. F never writes back to P or E, topologically blocking fault propagation.

03 · Radiation Hardening & ZL-0

Four-domain isolation and nanosecond-scale arbitration on discrete parts provide a new analytical framework for survival under regional radiation faults. The ZL-0 ground-potential-drift model is mathematically isomorphic to threshold-voltage drift under irradiation, making it transferable to radiation-hardened reliability analysis.

3.1 Versus conventional radiation-hardening

Conventional	Limitation	PEF Gate advantage
TMR (triple modular redundancy)	Regionally, radiation can defeat the physical	Attitude physically isolated from the main loop; regional faults cannot reach
Process hardening (SOI/SOS)	High cost, long cycle	Implemented with commercial off-the-shelf discrete parts; BOM cost at the
ECC error correction	Correction window exists	No memory, no bit-flip issue; fault implies immediate cut
Shield box	Heavy, volume-constrained	No extra shielding required; the architecture itself is radiation-survivable

3.2 ZL-0 ground-potential-drift model

The E-domain comparator threshold is generated by a precision resistor divider from the monitor rail. Tying the potentiometer ground to the monitor ground places the sense loop and the reference in the same thermodynamic reference frame, so the threshold is uniquely determined by the precision divider — eliminating drift from common-ground impedance.

This model is isomorphic to threshold drift under irradiation and can be migrated into quantitative radiation-reliability analysis. The isomorphism is the theoretical contribution — not the specific component values, which are restricted implementation details.

04 · Verification & Academic Value

4.1 Verification results

Metric	Result
Formal audit	ALPHA PASS — five-layer evidence chain closed
Arbiter latency	Nanosecond-scale typical; within sub-100ns budget at 85°C
Acceptance tests	All defined tests passed; engineering constraints hardened
Component-substitution red lines	Defined — critical components cannot be substituted without re-verification

4.2 Academic collaboration value

- Reproducible platform:** Provides a desktop-scale reproducible testbed for "safety layer works after main controller fails."
- Paper material:** ZL-0 drift model, discrete nanosecond arbitration, and physically isolated redundancy each yield independent papers.
- Grant leverage:** Offers a new technical path and innovation point for radiation-hardening grant proposals.

Four-domain physical isolation and a pure-hardware arbitration path ensure that no software error, processor fault, single-event upset, or EMP can prevent the safety boundary from opening the actuator cutoff before a physical boundary is breached.

— PEF Gate conclusion

05 · Compliance & Scope

This white paper publishes only the architecture definition, interface principles, and verified quantitative conclusions.

Restricted materials (not in this document): Complete schematics, BOM details, divider-threshold derivations, layout timing constraints, and production processes. These are released only to contracted collaborators.

The architecture is subject to the Export Administration Law of China and applicable regulations on dual-use items. Transfer of technical materials to foreign entities or provision of technical support to foreign subjects requires prior authorization.

PEF Gate · Public Release V1.0 · 2026-08

Architecture definition, interface principles, and verified quantitative conclusions only.
Complete schematics, BOM, derivations, and processes are restricted technical materials.
For technical collaboration and joint academic research, contact via official channels.